MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION (2024 ADMISSION ONWARDS)

MG4DSCECT201 – AI For Smart Electronics

Duration: 1.5 Hours Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall *outcome attainability*

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A – Multiple Choice Questions

Answer All Questions. Each question carries 1 mark

1. Which of the following best defines Artificial Intelligence? (U CO1)

a) Study of hardware systems

b) Simulation of human intelligence by machines

c) Internet-based programming

d) Database management

2. Which type of AI is designed for a specific task? (K CO1)

a) General AI b) Super AI

c) Narrow AI d) Strong AI

3. ML stands for: (K CO2)

a) Machine Logic b) Machine Learning

c) Modern Language d) Memory Load

4. Which learning technique uses labeled data? (K CO2)

a) Unsupervised Learning b) Reinforcement Learning

c) Supervised Learning d) Deep Learning

5. The basic unit of a neural network is: (K CO2)

a) Logic gate b) Artificial neuron

c) Register d) Sensor

6. Which layer receives input data in a neural network? (K CO2)

a) Hidden layer b) Output layer

c) Input layer d) Processing layer

7. Which of the following is an example of AI application? (U CO1)

a) Calculator b) Word Processor

c) Virtual Assistant d) Spreadsheet

8. Which hardware component is mainly used for parallel processing in AI? (K CO3)

a) RAM b) Hard Disk

c) GPU d) ROM

9. Face recognition belongs to which AI field? (K CO3)

a) Expert System b) Robotics

c) Computer Vision d) Database System

10. NLP stands for: (K CO3)

a) Natural Logic Program b) Neural Learning Process

c) Natural Language Processing d) Network Layer Protocol

1 × 10 = 10 Marks

Part B – Short Answer Questions

Answer FOUR Questions. Each question carries 5 marks

11. Explain the significance of Artificial Intelligence in modern society. (U CO1)
12. Differentiate between Narrow AI and General AI. (U CO1)
13. Explain Supervised Learning with an example. (U CO2)
14. Describe the structure of an artificial neuron with a neat diagram. (U CO2)
15. Write short notes on CPU and GPU selection for AI applications. (U CO3)
16. Explain any five applications of AI in everyday life. (A CO1)

4 × 5 = 20 Marks

Part C – Essay Questions

Answer any TWO Questions. Each question carries 10 marks

17. Explain the basics of Artificial Intelligence, its history, and importance in modern society. (U CO1)
18. Describe the different types of Artificial Intelligence: Narrow AI, General AI, and Generative AI with examples. (U CO1)
19. Explain Machine Learning and its types: Supervised, Unsupervised, and Reinforcement Learning with suitable examples. (U CO2)
20. Explain real-world applications of AI such as Virtual Assistants, Smart Home Automation, and Self-driving Cars. (U CO1)

2 × 10 = 20 Marks